Right Triangles & Trigonometry Error Analysis

TEACHER KEY Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Pythagorean theorem

ORIGINAL PROBLEM

A right triangle has legs 6 and 8. Find the hypotenuse.

INCORRECT WORKED SOLUTION

$c^2 = 6^2 + 8^2$

$c^2 = 100$

$c = 50$

A. EXPLAIN THE MISTAKE (1 POINT)

To undo the square, take the SQUARE ROOT of 100. Do not divide by 2. $c = \sqrt{100} = 10$.

B. SHOW CORRECT WORK (1 POINT)

$c^2 = 6^2 + 8^2$

$c^2 = 100$

$c = \sqrt{100} = 10$

2. Finding a leg

ORIGINAL PROBLEM

A right triangle has hypotenuse 13 and one leg 5. Find the other leg.

INCORRECT WORKED SOLUTION

$a^2 = 13^2 + 5^2$

$a^2 = 194$

$a \approx 13.9$

A. EXPLAIN THE MISTAKE (1 POINT)

The hypotenuse is the LONGEST side, so $a^2 + 5^2 = 13^2$. Subtract: $a^2 = 169 - 25 = 144$, so $a = 12$.

B. SHOW CORRECT WORK (1 POINT)

$a^2 + 5^2 = 13^2$

$a^2 = 169 - 25 = 144$

$a = 12$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Converse of the theorem

ORIGINAL PROBLEM

Is a triangle with sides 8, 15 and 17 a right triangle?

INCORRECT WORKED SOLUTION

$8 + 15 = 23$

$23 \neq 17$

It is not a right triangle.

A. EXPLAIN THE MISTAKE (1 POINT)

Use SQUARES of the sides: $8^2 + 15^2 = 64 + 225 = 289 = 17^2$. The equation holds, so it IS a right triangle.

B. SHOW CORRECT WORK (1 POINT)

$8^2 + 15^2 = 64 + 225 = 289$

$17^2 = 289$

They are equal, so it is a right triangle.

4. Trig ratio

ORIGINAL PROBLEM

In a right triangle, the side opposite ∠A is 5, the adjacent side is 12 and the hypotenuse is 13. Find sin A.

INCORRECT WORKED SOLUTION

$$\sin A = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\sin A = \frac{12}{13}$$

A. EXPLAIN THE MISTAKE (1 POINT)

Sine is OPPOSITE over hypotenuse (SOH). The wrong work used the adjacent side, which is cosine.

B. SHOW CORRECT WORK (1 POINT)

$$\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\sin A = \frac{5}{13}$$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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5. Solving for a side

ORIGINAL PROBLEM

∠A = 35°. The side adjacent to ∠A is 20. Find the opposite side x.

INCORRECT WORKED SOLUTION

$\tan 35° = \frac{20}{x}$

$x = \frac{20}{\tan 35°}$

$x \approx 28.6$

A. EXPLAIN THE MISTAKE (1 POINT)

Tangent is $\frac{\text{opposite}}{\text{adjacent}}$, so $\tan 35° = \frac{x}{20}$. Multiply: $x = 20 \cdot \tan 35° \approx 14.0$.

B. SHOW CORRECT WORK (1 POINT)

$\tan 35° = \frac{x}{20}$

$x = 20 \cdot \tan 35°$

$x \approx 14.0$

6. Finding an angle

ORIGINAL PROBLEM

In a right triangle, the side opposite ∠θ is 7 and the hypotenuse is 10. Find θ.

INCORRECT WORKED SOLUTION

$\cos \theta = \frac{7}{10} = 0.7$

$\theta = \cos^{-1}(0.7)$

$\theta \approx 45.6°$

A. EXPLAIN THE MISTAKE (1 POINT)

Opposite and hypotenuse go with SINE: $\sin \theta = \frac{7}{10}$, so $\theta = \sin^{-1}(0.7) \approx 44.4°$.

B. SHOW CORRECT WORK (1 POINT)

$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{7}{10}$

$\theta = \sin^{-1}(0.7)$

$\theta \approx 44.4°$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Special right triangle

ORIGINAL PROBLEM

A 45°-45°-90° triangle has legs of length 6. Find the hypotenuse.

INCORRECT WORKED SOLUTION

hypotenuse = leg · $\sqrt{3}$

hypotenuse = $6\sqrt{3}$

A. EXPLAIN THE MISTAKE (1 POINT)

In a 45°-45°-90° triangle, hypotenuse = leg · √2. (The √3 belongs to a 30°-60°-90° triangle.)

B. SHOW CORRECT WORK (1 POINT)

hypotenuse = leg · $\sqrt{2}$

hypotenuse = $6\sqrt{2} \approx 8.49$

8. Special right triangle

ORIGINAL PROBLEM

In a 30°-60°-90° triangle the shortest leg is 5. Find the hypotenuse.

INCORRECT WORKED SOLUTION

hypotenuse = short leg · $\sqrt{3}$

hypotenuse = $5\sqrt{3}$

A. EXPLAIN THE MISTAKE (1 POINT)

The hypotenuse is TWICE the short leg: 10. The √3 multiplier gives the LONG leg, 5√3.

B. SHOW CORRECT WORK (1 POINT)

hypotenuse = 2 · short leg = 10

long leg = $5\sqrt{3}$

Check: $5^2 + (5\sqrt{3})^2 = 100 = 10^2$ ✓

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9. Ladder problem

ORIGINAL PROBLEM

A 15-ft ladder leans on a wall with its base 5 ft from the wall. How high up the wall does it reach?

INCORRECT WORKED SOLUTION

$h^2 = 15^2 + 5^2$

$h^2 = 250$

$h \approx 15.8$ ft

A. EXPLAIN THE MISTAKE (1 POINT)

The ladder is the HYPOTENUSE, so $h^2 + 5^2 = 15^2$. The height must be less than 15 ft: $h^2 = 225 - 25 = 200$, so $h \approx 14.1$ ft.

B. SHOW CORRECT WORK (1 POINT)

$h^2 + 5^2 = 15^2$

$h^2 = 225 - 25 = 200$

$h = \sqrt{200} \approx 14.1$ ft

10. Angle of depression

ORIGINAL PROBLEM

From the top of a 50-ft tower, the angle of depression to a boat is 25°. How far is the boat from the base of the tower?

INCORRECT WORKED SOLUTION

$\tan 25° = \frac{d}{50}$

$d = 50 \cdot \tan 25°$

$d \approx 23.3$ ft

A. EXPLAIN THE MISTAKE (1 POINT)

The angle at the boat is also 25° (alternate interior angles). There, 50 is the OPPOSITE side and d is the ADJACENT side: $\tan 25° = \frac{50}{d}$, so $d = \frac{50}{\tan 25°} \approx 107.2$ ft.

B. SHOW CORRECT WORK (1 POINT)

Angle at the boat = 25° (alternate interior angles)

$\tan 25° = \frac{50}{d}$

$d = \frac{50}{\tan 25°} \approx 107.2$ ft

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points